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O Level · Cambridge · Additional Maths

1 hour 13 questions

Functions

Language of Functions / Inverse Functions / Composite Functions

Medium (5 questions)	/24
Hard (5 questions)	/27
Very Hard (3 questions)	/26
Total Marks	**/77**

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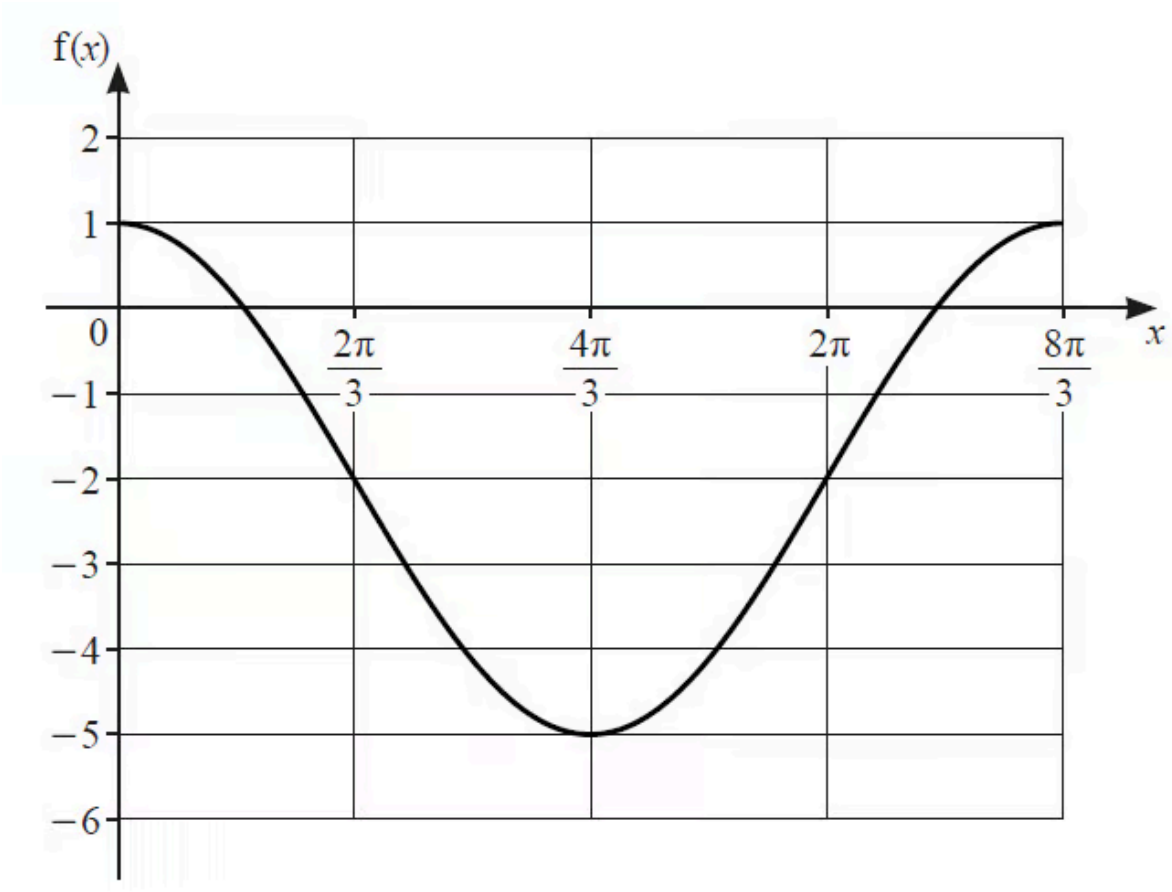
Medium Questions



1 (a) The diagram shows the graph of $f(x) = a\cos bx + c$ for $0 \le x \le \frac{8\pi}{3}$ radians.

Explain why f is a function.

(1 mark)

(b) Write down the range of f.

(1 mark)

2 $g(x) = 3 + \frac{1}{x}$ for $x \geq 1$.

(i) Find an expression for $g^{-1}(x)$.

[2]

(ii) Write down the range of g^{-1}.

[1]

(iii) Find the domain of g^{-1}.

[2]

(5 marks)

3 (a) It is given that $\mathrm{f}(x) = 5\ \ln(2x+3)$ for $x > -\frac{3}{2}$.

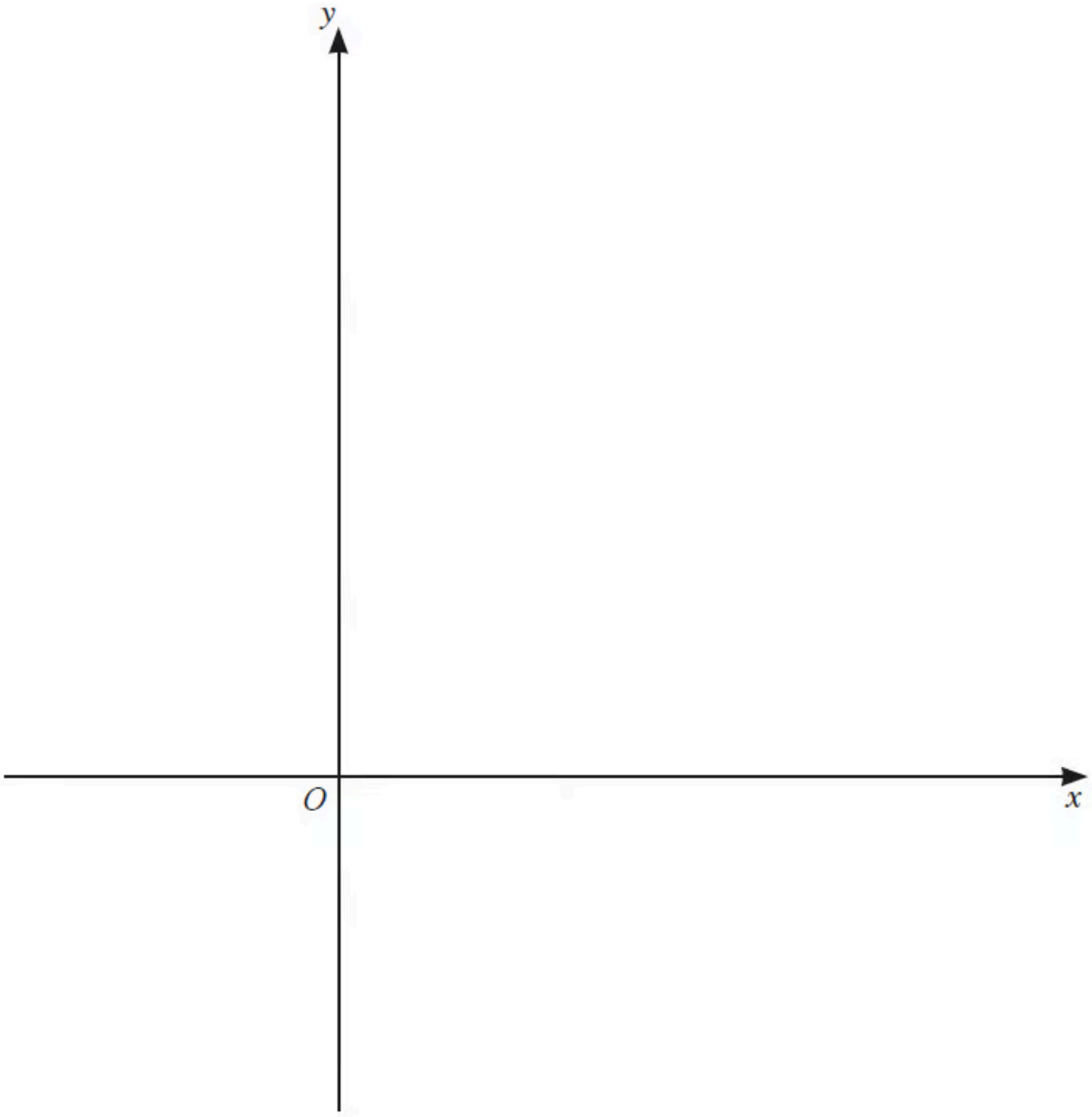
Write down the range of f.

(1 mark)

(b) Find f^{-1} and state its domain.

(3 marks)

(c) On the axes below, sketch the graph of $y = \mathrm{f}(x)$ and the graph of $y = \mathrm{f}^{-1}(x)$. Label each curve and state the intercepts on the coordinate axes.



(5 marks)

4 $$f(x) = 4\ln(2x-1)$$

(i) Write down the largest possible domain for the function f.

[1]

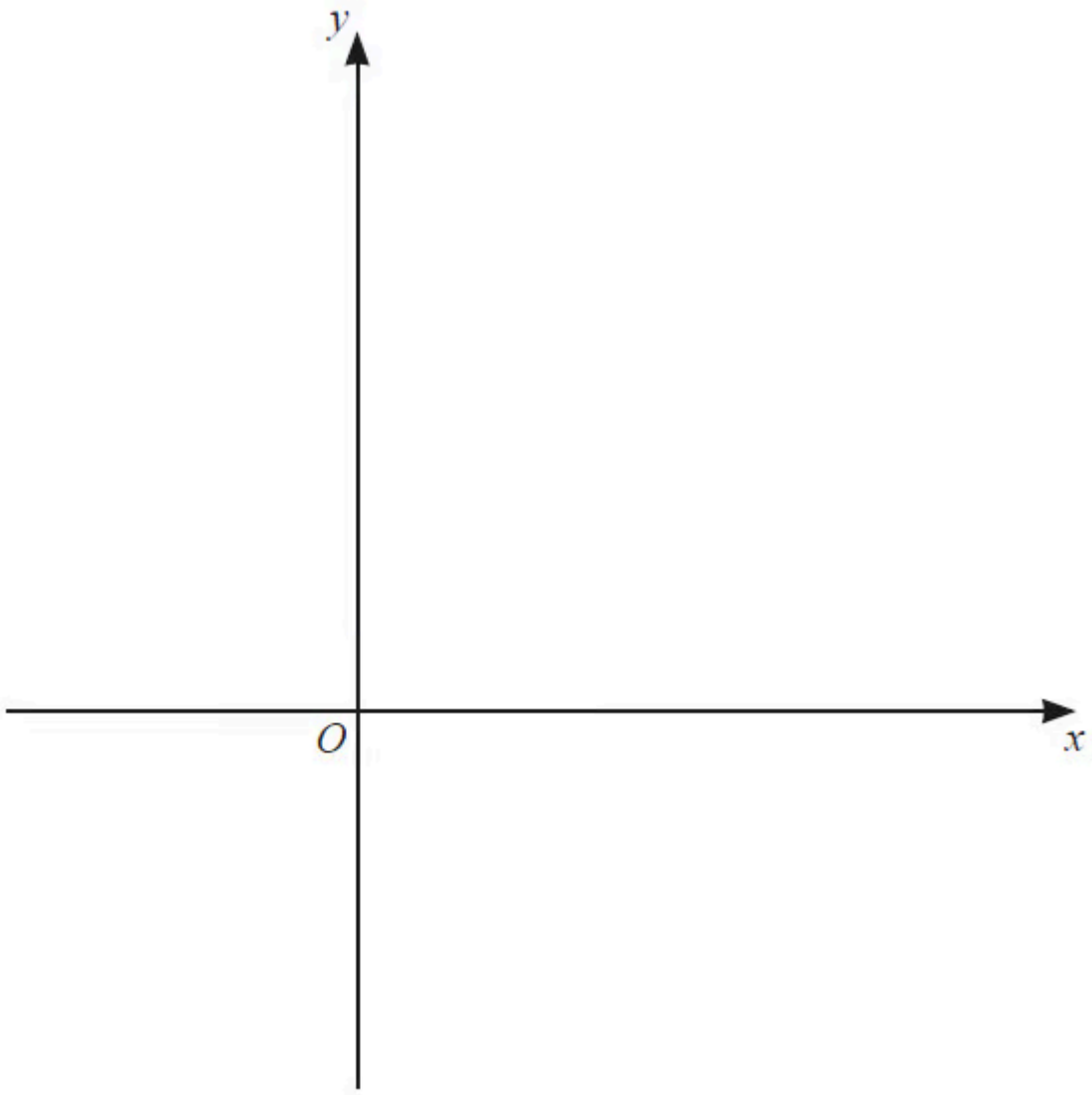
(ii) Find $\mathrm{f}^{-1}(x)$ and its domain.

[3]

(4 marks)

5 $\mathrm{h}(x) = 2\ln(3x-1)$ for $x \geq \frac{2}{3}$.

The graph of $y = \mathrm{h}(x)$ intersects the line $y = x$ at two distinct points. On the axes below, sketch the graph of $y = \mathrm{h}(x)$ and hence sketch the graph of $y = \mathrm{h}^{-1}(x)$.



(4 marks)

Hard Questions

1 It is given that $\mathrm{h}(x) = a + \dfrac{b}{x^2}$, where a and b are non-zero constants.

(i) Explain why $-2 \leqslant x \leqslant 2$ is not a suitable domain for $\mathrm{h}(x)$.

(ii) Given that $\mathrm{h}(1) = 4$ and $\mathrm{h}'(1) = 16$, find the values of a and b.

(3 marks)

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2 (a) $$\mathrm{f} : x \mapsto (2x+3)^2 \text{ for } x > 0$$

Find the range of f.

(1 mark)

(b) Explain why f has an inverse.

(1 mark)

(c) Find f^{-1}.

(3 marks)

(d) State the domain of f^{-1}.

(1 mark)

(e) Given that $\mathrm{g} : x \mapsto \ln(x+4)$ for $x > 0$, find the exact solution of $\mathrm{fg}(x) = 49$.

(3 marks)

3 $$\mathrm{g}(x) = x+5 \text{ for } x \in \mathbb{R}$$
$$\mathrm{h}(x) = \sqrt{2x-3} \text{ for } x \geq \frac{3}{2}$$

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Solve $\mathrm{gh}(x) = 7$.

(3 marks)

4 (a) The functions f and g are defined as follows.

$$\mathrm{f}(x) = x^2 + 4x \text{ for } x \in R$$
$$\mathrm{g}(x) = 1 + \mathrm{e}^{2x} \text{ for } x \in R$$

Find the range of f.

(2 marks)

(b) Write down the range of g.

(1 mark)

(c) Find the exact solution of the equation $\mathrm{fg}(x) = 21$, giving your answer as a single logarithm

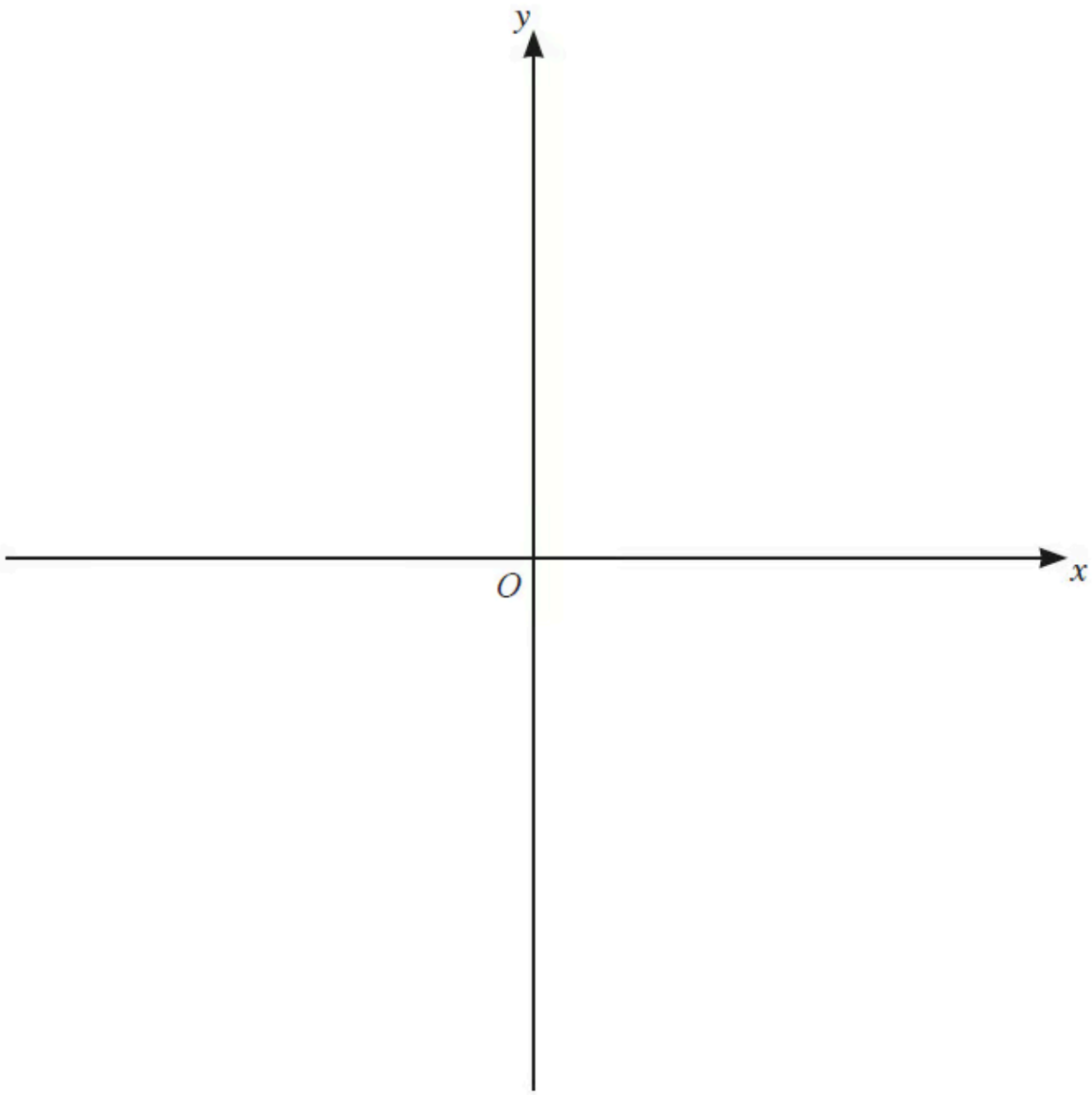
(4 marks)

5 (a) $$f(x) = x^2 + 2x - 3 \text{ for } x \geq -1$$

Given that the minimum value of $x^2 + 2x - 3$ occurs when $x = -1$, explain why $f(x)$ has an inverse.

(1 mark)

(b) On the axes below, sketch the graph of $y = f(x)$ and the graph of $y = f^{-1}(x)$. Label each graph and state the intercepts on the coordinate axes.



(4 marks)

Very Hard Questions

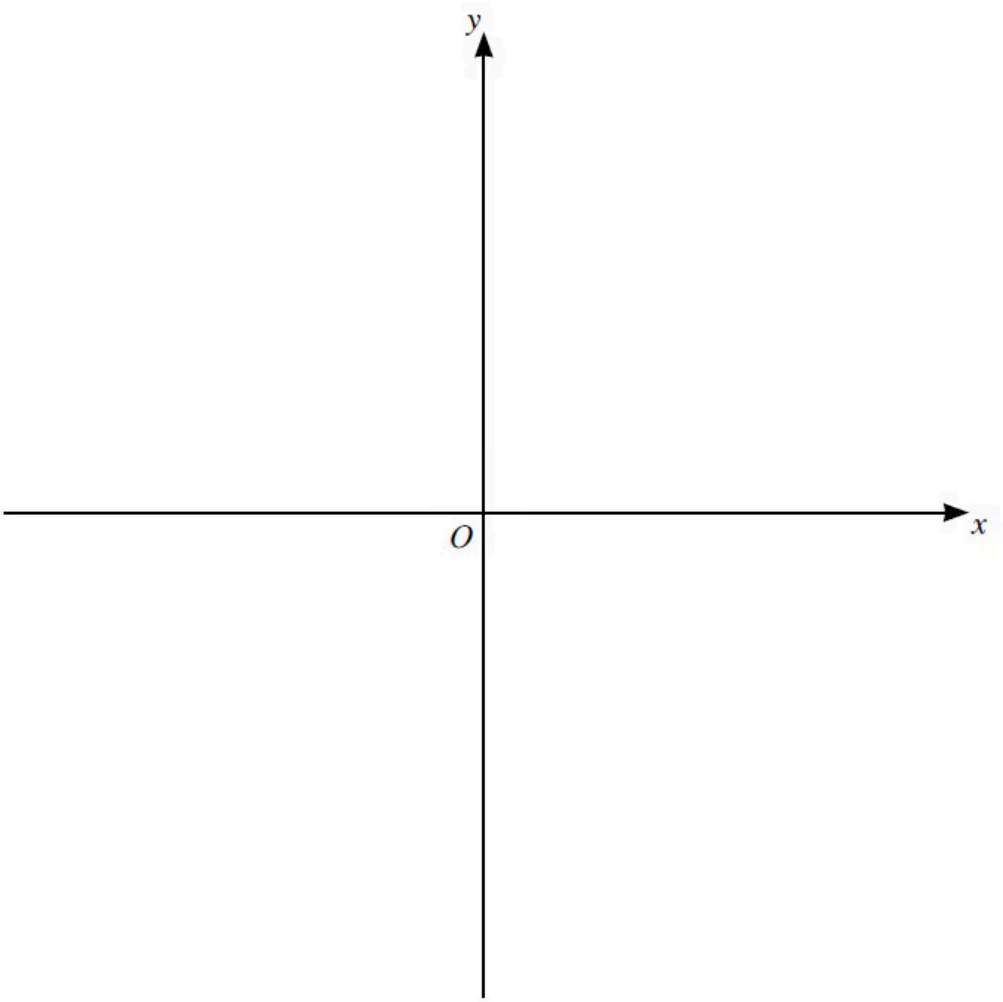
1 $f(x) = 3e^{2x} + 1$ for $x \in \mathbb{R}$
$g(x) = x + 1$ for $x \in \mathbb{R}$

(i) Write down the range of f and the range of g.

(ii) Find $g^2(0)$.

(iii) Hence find $fg^2(0)$.

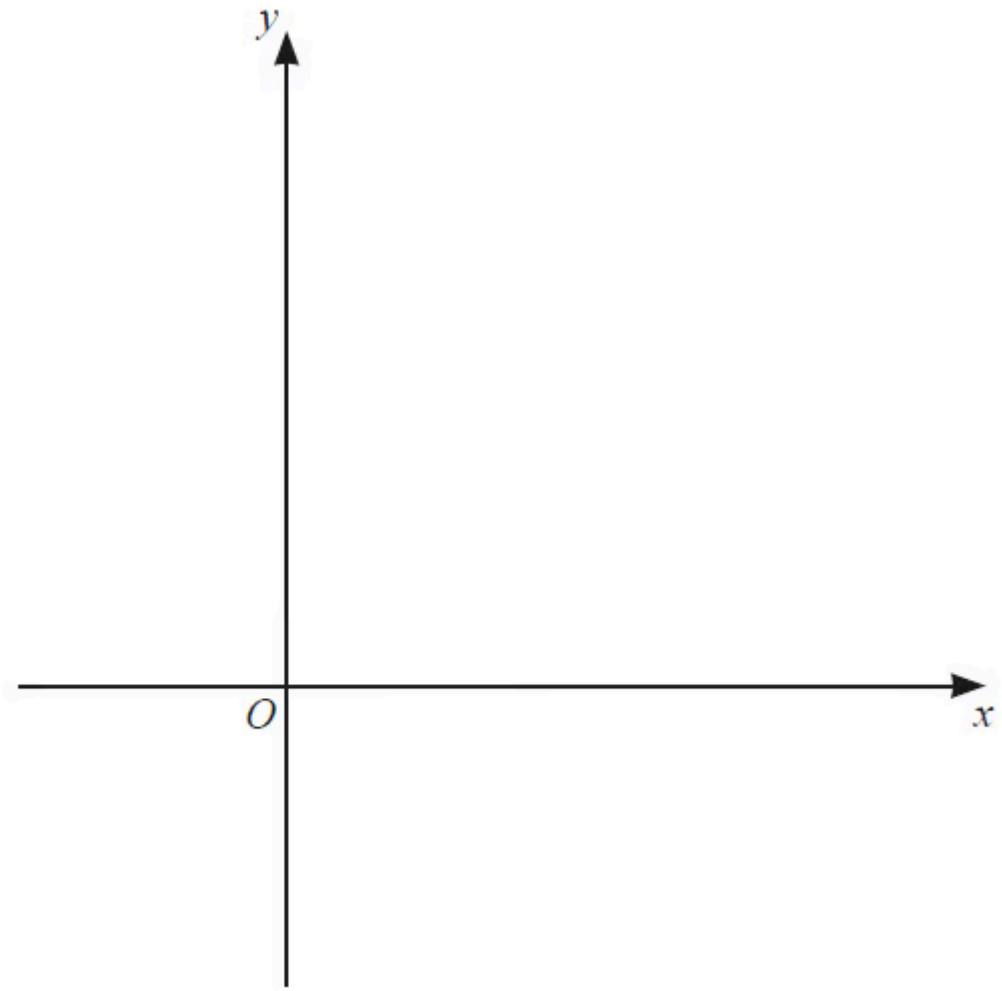
(iv) On the axes below, sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$.
State the intercepts with the coordinate axes and the equations of any asymptotes.



(9 marks)

2 (a) The function f is defined by $\mathrm{f}(x) = \ln(2x+1)$ for $x \geq 0$.

Sketch the graph of $y = \mathrm{f}(x)$ and hence sketch the graph of $y = \mathrm{f}^{-1}(x)$ on the axes below.



(3 marks)

(b) The function g is defined by $\mathrm{g}(x) = (x-4)^2 + 1$ for $x \leq 4$.

(i) Find an expression for $g^{-1}(x)$ and state its domain and range.

[4]

(ii) Find and simplify an expression for $fg(x)$.

[2]

(iii) Explain why the function gf does not exist.

[1]

(7 marks)

3 (a)

$$f(x) = 3 + e^x \text{ for } x \in \mathbb{R}$$
$$g(x) = 9x - 5 \text{ for } x \in \mathbb{R}$$

Find the range of f and of g.

(2 marks)

(b) Find the exact solution of $f^{-1}(x) = g'(x)$.

(3 marks)

(c) Find the solution of $g^2(x) = 112$.

(2 marks)

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